

When and how GLMs fail

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Complex data in the social and health sciences

so-sta.github.io/Materials



One question will guide the whole lecture

PASSI research question

How is the probability of having at **least one chronic disease** associated with age, sex, education and economic difficulties among adults interviewed in PASSI?

The response is binary:

$$Y_i = \begin{cases} 1, & \text{respondent } i \text{ declares at least one chronic disease,} \\ 0, & \text{otherwise.} \end{cases}$$

⚠ The numerical values used in worked examples can differ from the ones coming from your dataset.

Brief recap about GLMs

Logistic regression

What breaks when observations are dependent?

Brief recap about GLMs



Binary means require a bounded model

For a **Gaussian** linear model,

$$\mathbb{E}(Y_i | \mathbf{x}_i) = \mathbf{x}_i^\top \boldsymbol{\beta} \quad \text{can take any value in } \mathbb{R}.$$

For a **binary** response,

$$\mathbb{E}(Y_i | \mathbf{x}_i) = \Pr(Y_i = 1 | \mathbf{x}_i) = \pi_i \quad \text{must lie in } [0, 1].$$

The modeling problem

We want covariates to enter through a simple linear predictor, while the fitted mean respects the support and variance of the response.

A GLM has three connected components

1. Random component

Specifies the conditional distribution of Y_i and its mean–variance relationship:

$$\mathbb{E}(Y_i | \mathbf{x}_i) = \mu_i, \quad \text{Var}(Y_i | \mathbf{x}_i) = \underbrace{\phi}_{\text{dispersion}} \underbrace{V(\mu_i)}_{\text{mean–variance relationship}}.$$

2. Systematic component

Covariates enter through the linear predictor:

$$\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}.$$

3. Link function

Connects the conditional mean to the linear predictor:

$$g(\mu_i) = \eta_i.$$

For Bernoulli and Poisson GLMs, the dispersion parameter is fixed at $\phi = 1$.

The response distribution determines the variance function

Response	Distribution	$V(\mu)$	Typical link
Continuous	Gaussian	1	identity
Count	Poisson	μ	log
Binary	Bernoulli	$\mu(1 - \mu)$	logit
Positive continuous	Gamma	μ^2	log or inverse

The distribution **determines** $V(\mu)$, whereas the link function is a **modelling choice**.

Unlike a linear model, a GLM allows the conditional variance to change with the mean.

For the variable chronic disease, variability is largest near $\pi = 0.5$ and vanishes as π approaches 0 or 1.

The link chooses the scale on which effects add

$$g(\mu_i) = \eta_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}.$$

- **On the link scale:** a one-unit increase in x_r changes the linear predictor by β_r , holding the other covariates fixed:

$$\eta_i(x_r + 1) - \eta_i(x_r) = \beta_r.$$

- **On the response scale:** the fitted mean is obtained through the inverse link:

$$\mu_i = g^{-1}(\eta_i).$$

Therefore, the effect of a one-unit change in x_r is generally **not additive** and may depend on the current value of the covariates.

The link is not a transformation of Y_i : it transforms $\mu_i = \mathbb{E}(Y_i)$.

The exponential family supplies a common likelihood theory

Many GLM distributions can be written as

$$f(y_i; \theta_i, \phi) = \exp \left\{ \frac{y_i \theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi) \right\},$$

where θ_i is the **natural parameter**.

$$\mathbb{E}(Y_i) = b'(\theta_i), \quad \text{Var}(Y_i) = a(\phi) b''(\theta_i).$$

- Since $\mu_i = b'(\theta_i)$ the natural parameter can be written as $\theta_i = (b')^{-1}(\mu_i)$.
- The **canonical link** is defined as $\mathbf{g}_{\text{can}}(\mu_i) = (b')^{-1}(\mu_i)$, so that $\eta_i = \mathbf{g}_{\text{can}}(\mu_i) = \theta_i$.

Bernoulli data

$$\theta_i = \log \left(\frac{\mu_i}{1 - \mu_i} \right),$$

hence the canonical link is the **logit**.

Before fitting, four choices must be explicit

1. What distribution matches the **support** of the response?
2. Which **link function** connects the mean to the linear predictor?
3. Which covariates and functional forms define the **mean model**?
4. Which observations can reasonably be treated as **independent**?

Logistic regression



PASSI provides a genuine Bernoulli outcome

$$Y_i = \begin{cases} 1, & \text{respondent } i \text{ declares at least one chronic disease,} \\ 0, & \text{otherwise.} \end{cases}$$

For respondent i ,

$$Y_i \mid \mathbf{x}_i \sim \text{Bernoulli}(\pi_i), \quad \pi_i = \Pr(Y_i = 1 \mid \mathbf{x}_i).$$

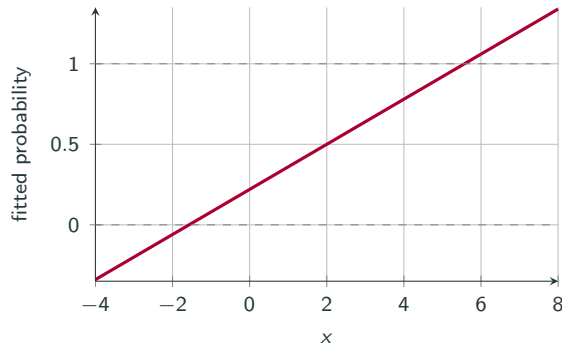
with **sex, age, educational level and economic difficulties** included in the **covariate** vector $\mathbf{x}_i$.

$$\Pr(Y_i = y_i \mid \mathbf{x}_i) = \pi_i^{y_i} (1 - \pi_i)^{1-y_i}, \quad y_i \in \{0, 1\}.$$

$$\mathbb{E}(Y_i \mid \mathbf{x}_i) = \pi_i, \quad \text{Var}(Y_i \mid \mathbf{x}_i) = \pi_i(1 - \pi_i).$$

- **The variance is determined by the mean:** unlike in the Gaussian linear model, there is no separate variance parameter σ^2 .
- **The variance changes with π_i :** it is largest at $\pi_i = 0.5$ and approaches zero as π_i approaches 0 or 1.
- Once the mean is specified, the marginal variance of each observation is fixed; **additional dependence or variability requires additional modelling.**
- **The mean is constrained to $(0, 1)$:** a suitable link function connects π_i to the unconstrained linear predictor $\eta_i \in \mathbb{R}$.

Linear probability model



$$\pi_i = \beta_0 + \beta_1 x_i$$

is easy to read, but it can give $\hat{\pi}_i < 0$ or $\hat{\pi}_i > 1$.

It assumes a **constant additive effect** of x on the probability scale.

Logit function

The **logit** maps probabilities to the whole real line

$$\text{logit}(\pi) = \log \left(\frac{\pi}{1 - \pi} \right) : (0, 1) \rightarrow \mathbb{R}$$

π	0.01	0.20	0.50	0.80
$\text{logit}(\pi)$	-4.60	-1.39	0	1.39

- Probabilities below 0.5 have negative logits.
- $\pi = 0.5$ corresponds to zero log-odds.
- The transformation is symmetric around 0.5.

Logit function

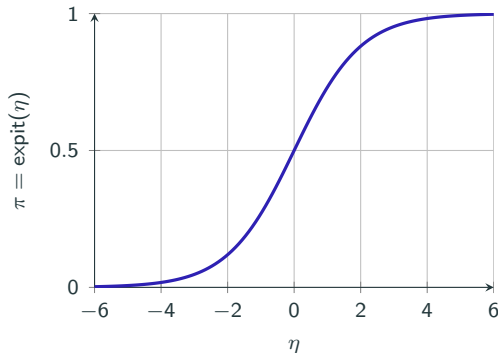
The **inverse link** returns valid probabilities

$$\pi_i = \text{expit}(\eta_i) = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}.$$

Whatever the value of $\eta_i \in \mathbb{R}$,

$$0 < \pi_i < 1.$$

The S-shape is almost linear near 0.5 and flatter near 0 and 1.



The **odds** are defined as

$$\text{odds}(\pi) = \frac{\pi}{1 - \pi}.$$

Illustration

Let $\pi = \Pr(Y = 1)$, where $Y = 1$ indicates having at least one chronic condition. If $\pi = 0.20$, then

$$\text{odds}(\pi) = \frac{\pi}{1 - \pi} = \frac{0.20}{0.80} = 0.25.$$

Thus, we expect approximately one person with at least one chronic condition for every four people without chronic conditions.

The **odds ratio** compares the odds in two groups:

$$OR = \frac{\text{odds}(\pi_1)}{\text{odds}(\pi_0)}.$$

Illustration

Let $X \in \{0, 1\}$ identify two groups and $\pi_1 = \Pr(Y = 1 \mid X = 1) = 0.40$, $\pi_0 = \Pr(Y = 1 \mid X = 0) = 0.20$. Therefore,

$$OR = \frac{\text{odds}(\pi_1)}{\text{odds}(\pi_0)} = \frac{0.67}{0.25} \approx 2.67.$$

The odds of having at least one chronic condition are approximately 2.7 times as high in group $X = 1$ as in the reference group $X = 0$.

Returning to the logistic model

For the logistic regression model

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 x_i,$$

a **one-unit increase** in x gives

$$\log \left(\frac{\text{odds}(\pi(x+1))}{\text{odds}(\pi(x))} \right) = \beta_1,$$

and therefore

$$\boxed{\text{OR} = e^{\beta_1}}.$$

Illustration

If $\beta_1 = \log(2)$, then

$$\text{OR} = e^{\beta_1} = 2.$$

A one-unit increase in x doubles the odds of the outcome.

PASSI example

For individual i , let age_i denote age in years, and let

$$\text{sex}_i = \begin{cases} 1 & \text{if individual } i \text{ is male,} \\ 0 & \text{if individual } i \text{ is female,} \end{cases} \quad \text{education}_i = \begin{cases} 1 & \text{if individual } i \text{ has low education,} \\ 0 & \text{otherwise,} \end{cases}$$

$$\text{economic}_i = \begin{cases} 1 & \text{if individual } i \text{ reports economic difficulties,} \\ 0 & \text{otherwise.} \end{cases}$$

Thus, the covariate vector is

$$\mathbf{x}_i = (1, \text{age}_i, \text{sex}_i, \text{education}_i, \text{economic}_i)^\top.$$

Let

$$\pi_i = \Pr(Y_i = 1 \mid \text{age}_i)$$

be the probability that respondent i reports at least one chronic disease.

We start with the simple logistic regression model

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 \text{age}_i$$

or, equivalently,

$$\pi_i = \frac{\exp(\beta_0 + \beta_1 \text{age}_i)}{1 + \exp(\beta_0 + \beta_1 \text{age}_i)}.$$

Interpreting the effect of age

Consider two individuals whose **ages** differ by one year:

$$\text{logit}\{\pi(\text{age} = a + 1)\} - \text{logit}\{\pi(\text{age} = a)\} = \beta_1.$$

Exponentiating,

$$\frac{\text{odds}(Y = 1 \mid \text{age} = a + 1)}{\text{odds}(Y = 1 \mid \text{age} = a)} = \exp(\beta_1).$$

- β_1 is the change in **log-odds** associated with a one-year increase in **age**.
- $\exp(\beta_1)$ is the **odds ratio associated with a one-year increase in age**.
- The odds ratio is constant across **age**, but the corresponding difference in probability is **not constant**.

Fitting the age model in R

```
> m_age <- glm(chronic ~ age, data = passi, family = binomial(link = "logit"))

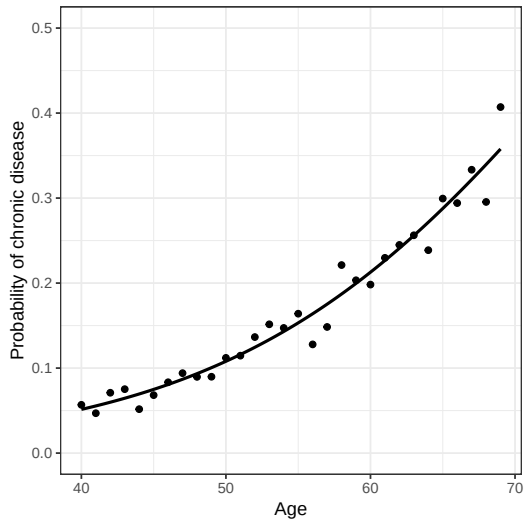
> summary(m_age)
Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -6.128624    0.203130  -30.17  <2e-16 ***
age          0.080341    0.003461   23.21  <2e-16 ***

> exp(coef(m_age)) #odds ratio
(Intercept)          age
0.002179579  1.083656668

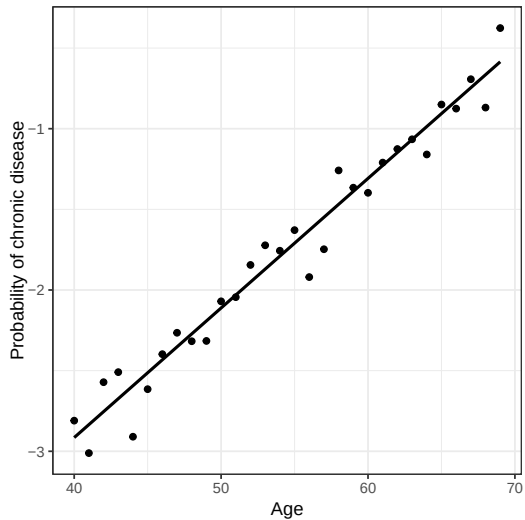
> exp(confint(m_age))["age",] #95% CI
2.5 %      97.5 %
age      1.076372484 1.091078854
```

➔ Each additional year of **age** is associated with approximately $(1.084-1) \times 100 = 8.4\%$ **higher odds** of reporting at least one chronic disease.

Probability versus age



Logit versus age



Adding a categorical predictor: sex

Now we extend the model to

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 \text{age}_i + \beta_2 \text{sex}_i$$

Females are the **reference category**.

Therefore,

$$\text{Females:} \quad \text{logit}(\pi_F) = \beta_0 + \beta_1 \text{age},$$

$$\text{Males:} \quad \text{logit}(\pi_M) = (\beta_0 + \beta_2) + \beta_1 \text{age}.$$

Interpreting age and sex

- β_1 measures the effect of a one-year increase in **age** holding **sex** fixed:

$$\text{OR}_{\text{age}} = \exp(\beta_1).$$

- β_2 compares males with females **holding age fixed**:

$$\text{logit}(\pi_M) - \text{logit}(\pi_F) = \beta_2,$$

and therefore

$$\boxed{\frac{\text{odds}_M}{\text{odds}_F} = \exp(\beta_2)}.$$

➔ $\exp(\beta_2)$ is the **age-adjusted odds ratio for males versus females**.

What does adding sex do geometrically?

Without interaction,

$$\text{Females: } \text{logit}(\pi_F) = \beta_0 + \beta_1 \text{ age},$$

$$\text{Males: } \text{logit}(\pi_M) = (\beta_0 + \beta_2) + \beta_1 \text{ age}.$$

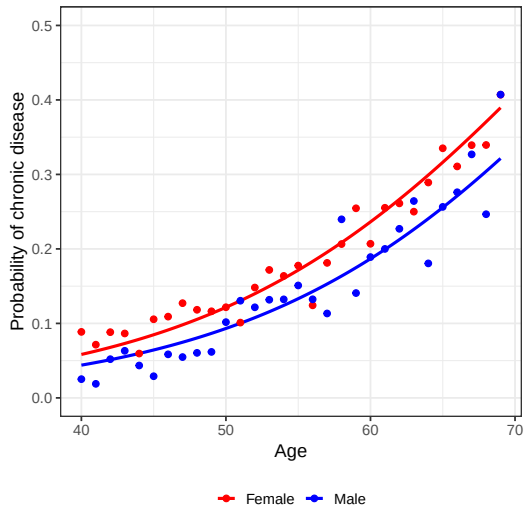
- Both groups have the **same slope** β_1 .
- Sex changes only the **level** of the linear predictor.
- On the log-odds scale, the two fitted lines are therefore **parallel**.
- On the probability scale, the two logistic curves are shifted, but the vertical difference between probabilities is **not constant across age**.

Age + sex in R

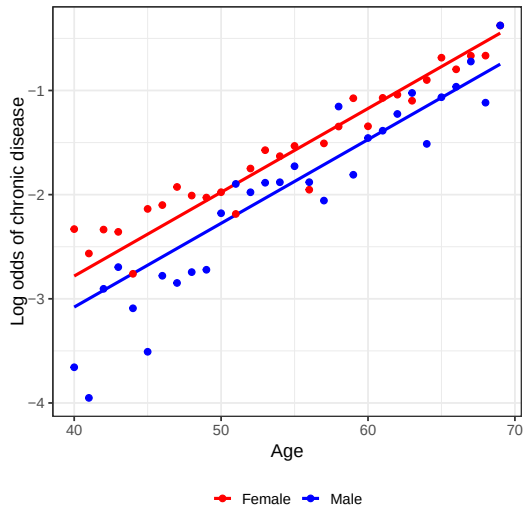
```
> m_age_sex <- glm(chronic ~ age + sex, data = passi, family = binomial(link =  
  "logit"))  
  
> summary(m_age_sex)  
  
Coefficients:  
                Estimate Std. Error z value Pr(>|z|)  
(Intercept) -5.998540    0.204508 -29.332  < 2e-16 ***  
age          0.080425    0.003468  23.189  < 2e-16 ***  
sex         -0.297947    0.055892  -5.331  9.78e-08 ***  
  
> exp(coef(m_age_sex))  
(Intercept)          age          sex  
0.002482373 1.083747294 0.742340552
```

➔ At the same age, males have about **0.74 times the odds** of reporting at least one chronic disease compared with females. Equivalently, $(1 - 0.7423) \times 100 \approx 25.8\%$, so their odds are approximately **25.8% lower**.

Probabilities by sex



Logit by sex



Do males and females have the same age effect?

The previous model assumes

β_1 = the same **age** effect for males and females.

But this may be too restrictive.

We may instead ask:

Research question

Does the association between **age** and chronic disease differ between males and females?

To allow the **age** effect to depend on **sex**, we introduce an **interaction**:

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 \text{age}_i + \beta_2 \text{sex}_i + \beta_3 \text{age}_i \text{sex}_i$$

Age-by-sex interaction

For females, $\text{sex} = 0$:

$$\text{logit}(\pi_F) = \beta_0 + \beta_1 \text{age}.$$

For males, $\text{sex} = 1$:

$$\begin{aligned}\text{logit}(\pi_M) &= \beta_0 + \beta_1 \text{age} + \beta_2 + \beta_3 \text{age} \\ &= (\beta_0 + \beta_2) + (\beta_1 + \beta_3) \text{age}.\end{aligned}$$

Therefore, the interaction allows the two groups to have both **different levels** and **different slopes** on the log-odds scale.

How do we interpret the interaction coefficients?

- β_1 is the **age** effect for **females**
- $\beta_1 + \beta_3$ is the **age** effect for **males**
- β_3 measures how much the **age** effect differs between males and females on the log-odds scale. If $\beta_3 = 0$, the **age** effect is the same for males and females.
- Equivalently,

$$\exp(\beta_3) = \frac{\exp(\beta_1 + \beta_3)}{\exp(\beta_1)} = \frac{\text{OR}_{\text{age}|M}}{\text{OR}_{\text{age}|F}}.$$

A subtle point: the sex effect now depends on age

With an interaction,

$$\text{logit}(\pi_M) - \text{logit}(\pi_F) = \beta_2 + \beta_3 \text{age}.$$

Therefore,

$$\text{OR}_{M:F}(\text{age}) = \exp(\beta_2 + \beta_3 \text{age})$$

- There is **no** longer a single **odds ratio for males versus females**.
- The sex effect changes with age.
- In particular, $\exp(\beta_2)$ is the male–female odds ratio when **age** = 0.

⚠ If **age** zero is outside the observed range, β_2 by itself may have little practical interpretation.

Centering age makes the interaction easier to interpret

Choose a **meaningful reference age**, for example 50 years:

$$\text{age50}_i = \text{age}_i - 50.$$

Then fit

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 \text{age50}_i + \beta_2 \text{sex}_i + \beta_3 \text{age50}_i \text{sex}_i.$$

Now:

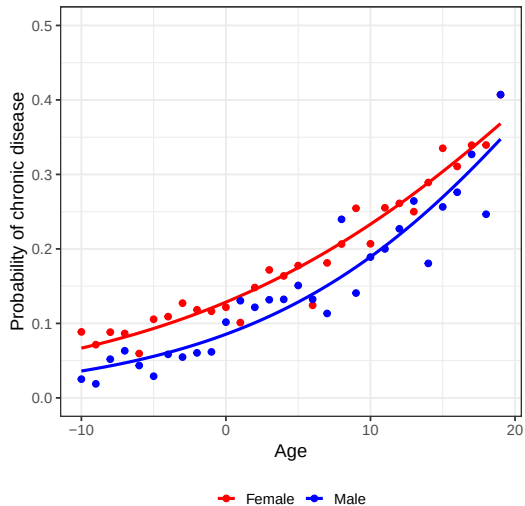
- β_0 is the log-odds for a **50-year-old female**;
- $\exp(\beta_2)$ is the male–female odds ratio **at age 50**;
- β_1 and $\beta_1 + \beta_3$ remain the age slopes for females and males, respectively.

Fitting the interaction in R

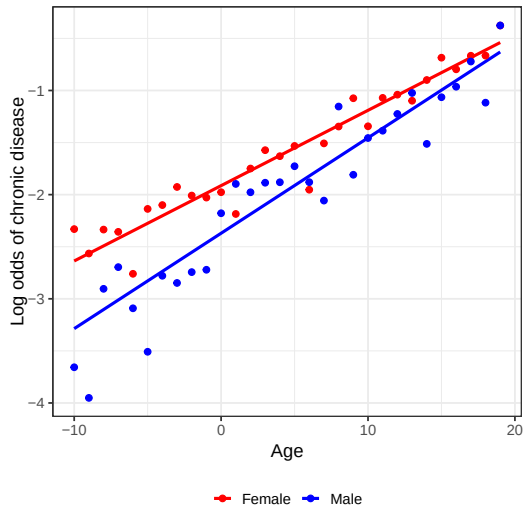
```
>m_int50 <- glm(chronic ~ age50 * sex, data = passi, family = binomial(link =  
  "logit"))  
  
>summary(m_int50)  
Coefficients:  
             Estimate Std. Error z value Pr(>|z|)  
(Intercept) -1.913673   0.050344 -38.012  < 2e-16 ***  
age50         0.072319   0.004512  16.027  < 2e-16 ***  
sex          -0.456859   0.081304  -5.619  1.92e-08 ***  
age50:sex      0.019271   0.007065   2.728  0.00637 **  
  
> exp(coef(m_int50))  
(Intercept)      age50      sex    age50:sex  
0.1475376    1.0749983    0.6332694    1.0194579
```

➔ Let's read our estimates!

Interaction model - Probability scale



Interaction model - Logit scale



Final model in R

```
> m_final <- glm(chronic ~ age50*sex + eco + edu, data = passi, family =  
  binomial(link = "logit"))  
  
> summary(m_final)  
Coefficients:  
                Estimate Std. Error z value Pr(>|z|)  
(Intercept) -1.940216    0.060292 -32.180  < 2e-16 ***  
age50         0.069811    0.004224  16.529  < 2e-16 ***  
sex          -0.544630    0.077622  -7.016  2.28e-12 ***  
eco           0.214705    0.053243   4.033  5.52e-05 ***  
edu           0.210304    0.053231   3.951  7.79e-05 ***  
age50:sex     0.023724    0.006815   3.481  0.000499 ***  
  
> exp(coef(m_final))  
(Intercept)      age50      sex      eco      edu  
0.1436729    1.0723060    0.5800562    1.2394957    1.2340535  
age50:sex  
1.0240079
```

Testing a regression effect

To test

$$H_0 : \beta_r = 0 \quad \text{vs} \quad H_1 : \beta_r \neq 0,$$

the Wald statistic is

$$Z_r = \frac{\hat{\beta}_r}{\text{SE}(\hat{\beta}_r)}.$$

Under H_0 and standard regularity conditions,

$$Z_r \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{as } n \rightarrow \infty,$$

where n is the number of independent observations.

How are $\hat{\beta}_r$ and $\text{SE}(\hat{\beta}_r)$ obtained?

Maximum likelihood estimation

For independent Bernoulli observations,

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^n [y_i \log\{\pi_i(\boldsymbol{\beta})\} + (1 - y_i) \log\{1 - \pi_i(\boldsymbol{\beta})\}].$$

The MLE **maximizes** $\ell(\boldsymbol{\beta})$ and, when a finite interior maximum exists, satisfies

$$\mathbf{U}(\hat{\boldsymbol{\beta}}) = \left. \frac{\partial \ell(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \right|_{\hat{\boldsymbol{\beta}}} = \mathbf{0}.$$

For logistic regression,

$$\mathbf{U}(\boldsymbol{\beta}) = \mathbf{X}^\top \{\mathbf{y} - \boldsymbol{\pi}(\boldsymbol{\beta})\}.$$

➔ There is **no closed-form solution**: software solves the score equations iteratively (Newton–Raphson/Fisher scoring; IRLS for logistic regression).

From the score to the variance of $\hat{\beta}$

Let β_0 denote the true parameter value. A first-order Taylor expansion of the score gives

$$\hat{\beta} - \beta_0 \approx \mathbf{A}_0^{-1} \mathbf{U}(\beta_0),$$

where

$$\mathbf{A}_0 = -\mathbb{E} \left[\frac{\partial \mathbf{U}(\beta)}{\partial \beta^\top} \bigg|_{\beta=\beta_0} \bigg| \mathbf{X} \right].$$

Therefore,

$$\boxed{\text{Var}(\hat{\beta} \mid \mathbf{X}) \approx \mathbf{A}_0^{-1} \mathbf{B}_0 \mathbf{A}_0^{-1}} \quad \text{with} \quad \mathbf{B}_0 = \text{Var}\{\mathbf{U}(\beta_0) \mid \mathbf{X}\}.$$

Two different quantities

$\mathbf{A}_0$ measures the **sensitivity** of the score to changes in β , whereas $\mathbf{B}_0$ measures **how much** the score actually varies.

Under independence: the information equality

For a correctly specified **independent** logistic likelihood,

$$\mathbf{A}_0 = \mathbf{B}_0 = \mathbf{X}^\top \mathbf{W}_0 \mathbf{X},$$

where

$$\mathbf{W}_0 = \text{diag} \{ \pi_i(\beta_0)[1 - \pi_i(\beta_0)] \}.$$

Hence,

$$\text{Var}(\hat{\beta} \mid \mathbf{X}) \approx \mathbf{A}_0^{-1} \mathbf{A}_0 \mathbf{A}_0^{-1} = \mathbf{A}_0^{-1},$$

so that

$$\boxed{\text{Var}(\hat{\beta} \mid \mathbf{X}) \approx (\mathbf{X}^\top \mathbf{W}_0 \mathbf{X})^{-1}}.$$

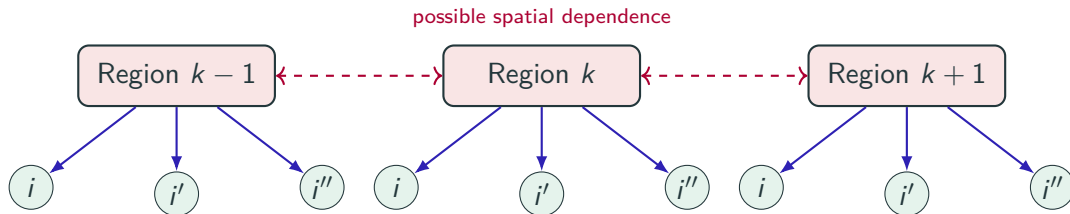
Replacing β_0 with $\hat{\beta}$ gives

$$\boxed{\widehat{\text{Var}}(\hat{\beta})_{\text{GLM}} = (\mathbf{X}^\top \widehat{\mathbf{W}} \mathbf{X})^{-1}}.$$

What breaks when observations are dependent?



PASSI observations are clustered within regions



- Individuals within the same **region** share contextual, social, environmental and health-system characteristics, and their outcomes may therefore be more similar.
 - Moreover, **nearby regions may themselves be similar**, generating spatial dependence across regions.
- ➔ The independence assumption may therefore fail both **within regions** and, potentially, **between neighboring regions**.

Let consider one level of aggregation, i.e., **region**.

Symbol	Meaning
$k = 1, \dots, K$	region
$i = 1, \dots, n_k$	respondent within region k
Y_{ik}	binary chronic disease indicator
$\mathbf{x}_{ik}$	individual and contextual covariates
π_{ik}	$\Pr(Y_{ik} = 1 \mid \mathbf{x}_{ik})$

Independence is conditional on the covariates

The ordinary logistic GLM assumes

$$Y_{ik} \perp\!\!\!\perp Y_{h\ell} \mid \mathbf{X} \quad \text{for all } (i, k) \neq (h, \ell),$$

where i indexes individuals and k regions.

This means that, once the observed covariates are taken into account, knowing **the outcome of one respondent should provide no additional information about another respondent.**

PASSI

This assumption may be unrealistic if respondents share unobserved regional characteristics or if geographically close regions are themselves similar.

Dependence appears through non-zero covariances

There are two possible sources of **dependence**:

1. **Within-region dependence**

For two different individuals $i \neq h$ in the same region k ,

$$\text{Cov}(Y_{ik}, Y_{hk} \mid \mathbf{X}) \neq 0.$$

They may be correlated because they share the same regional context.

2. **Between-region dependence**

For individuals living in different regions $k \neq \ell$,

$$\text{Cov}(Y_{ik}, Y_{h\ell} \mid \mathbf{X}) \neq 0$$

may arise when regions are spatially or contextually related.

By contrast, the ordinary GLM assumes $\text{Cov}(Y_{ik}, Y_{h\ell} \mid \mathbf{X}) = 0$ for every $(i, k) \neq (h, \ell)$.

Dependence changes the covariance of the responses

Consider $\mathbf{Y} = (Y_{1k} \ Y_{2k} \ Y_{1\ell} \ Y_{2\ell})^\top$. Under independence,

$$\text{Var}(\mathbf{Y} \mid \mathbf{X}) = \mathbf{W}_0.$$

With dependence, instead,

$$\text{Var}(\mathbf{Y} \mid \mathbf{X}) = \mathbf{\Omega}_0,$$

where

$$\mathbf{\Omega}_0 = \begin{pmatrix} \text{Var}(Y_{1k}) & \text{Cov}(Y_{1k}, Y_{2k}) & \text{Cov}(Y_{1k}, Y_{1\ell}) & \text{Cov}(Y_{1k}, Y_{2\ell}) \\ \text{Cov}(Y_{2k}, Y_{1k}) & \text{Var}(Y_{2k}) & \text{Cov}(Y_{2k}, Y_{1\ell}) & \text{Cov}(Y_{2k}, Y_{2\ell}) \\ \text{Cov}(Y_{1\ell}, Y_{1k}) & \text{Cov}(Y_{1\ell}, Y_{2k}) & \text{Var}(Y_{1\ell}) & \text{Cov}(Y_{1\ell}, Y_{2\ell}) \\ \text{Cov}(Y_{2\ell}, Y_{1k}) & \text{Cov}(Y_{2\ell}, Y_{2k}) & \text{Cov}(Y_{2\ell}, Y_{1\ell}) & \text{Var}(Y_{2\ell}) \end{pmatrix}.$$

➔ The ordinary GLM assumes $\mathbf{\Omega}_0 = \mathbf{W}_0$, i.e. all off-diagonal covariances are zero.

Which variance should we use for $\hat{\beta}$?

Suppose we continue to fit the **working-independence logistic GLM**, even though the observations are dependent.

Its score is still

$$\mathbf{U}(\beta) = \mathbf{X}^\top \{\mathbf{Y} - \pi(\beta)\}.$$

The sensitivity remains

$$\mathbf{A}_0 = -\mathbb{E} \left[\left. \frac{\partial \mathbf{U}(\beta)}{\partial \beta^\top} \right|_{\beta=\beta_0} \middle| \mathbf{X} \right] = \mathbf{X}^\top \mathbf{W}_0 \mathbf{X}.$$

However, the **actual score variance** is now

$$\mathbf{B}_0 = \text{Var}\{\mathbf{U}(\beta_0) \mid \mathbf{X}\} = \mathbf{X}^\top \mathbf{\Omega}_0 \mathbf{X}.$$

Therefore, under dependence $\Rightarrow \mathbf{B}_0 \neq \mathbf{A}_0$ in general.

The sandwich estimator

Let

$$\mathbf{A}_0 = \mathbf{X}^\top \mathbf{W}_0 \mathbf{X}, \quad \mathbf{B}_0 = \text{Var}\{\mathbf{U}(\beta_0) \mid \mathbf{X}\}.$$

Since $\mathbf{A}_0$ and $\mathbf{B}_0$ are unknown, we estimate them:

$$\widehat{\text{Var}}(\hat{\beta})_{\text{sand}} = \underbrace{\hat{\mathbf{A}}^{-1}}_{\text{bread}} \underbrace{\hat{\mathbf{B}}}_{\text{meat}} \underbrace{\hat{\mathbf{A}}^{-1}}_{\text{bread}}.$$



Bread

$\hat{\mathbf{A}}^{-1}$ describes how variability in the score is translated into variability of $\hat{\beta}$.

Meat

$\hat{\mathbf{B}}$ estimates the **actual variability of the score**, including the variability induced by dependence.

➔ Different robust covariance estimators mainly differ in **how the meat $\hat{\mathbf{B}}$ is estimated**.

Too-small standard errors produce overconfident inference

If $\hat{\beta}_r$ remains similar but its standard error is **underestimated**,

$$|Z_r| = \left| \frac{\hat{\beta}_r}{\text{SE}(\hat{\beta}_r)} \right| = \left| \frac{\hat{\beta}_r}{\sqrt{[\widehat{\text{Var}}(\hat{\beta})]_{rr}}} \right|$$

tends to be too large.

Consequences

- confidence intervals too narrow;
- p -values too small;
- **inflated Type I error**;
- apparent precision larger than the actual information.

Effective information

Many individuals from the same region do not provide the same amount of **independent information** as the same number of individuals drawn from independent contexts.

What may fail first is inference, not the coefficient

Suppose the marginal mean model is correctly specified:

$$\text{logit}(\pi_{ik}) = \mathbf{x}_{ik}^{\top} \beta.$$

Ignoring dependence does **not necessarily** imply that $\hat{\beta}$ is **biased**.

Under suitable conditions, estimating equations based on working independence can still target the correct marginal regression coefficients.

➔ But the usual GLM inference is no longer justified.

Moreover, if the **mean model** itself is **misspecified** because relevant regional structure is omitted, the coefficient estimates may also change.

Adding region indicators does not generally restore independence

One possibility is to include **region fixed effects**:

$$\text{logit}(\pi_{ik}) = \alpha_k + \mathbf{x}_{ik}^\top \beta.$$

This allows the average log-odds to differ across regions.

However, region indicators do not automatically remove

- **residual dependence** among individuals within a region;
- **unobserved regional mechanisms** not represented by a constant shift;
- **spatial dependence** between neighboring regions.

Mean adjustment $\neq$ covariance adjustment

Adding region indicators changes the **mean model**.

It does not, by itself, provide a valid model for the **covariance structure**.

Longitudinal data

	Longitudinal data	PASSI regional data
Observations	repeated measurements on the same person	different individuals within regions
Source of dependence	persistent individual traits and temporal dynamics	shared regional context
Possible additional structure	correlation may depend on time lag	correlation may depend on geographic proximity
Independent rows?	generally no	generally not guaranteed

What is the relevant dependence structure in PASSI?

Before choosing an inferential method, we should ask:

1. Are respondents within the same region plausibly correlated?
2. Can respondents from different but geographically close regions also be correlated?
3. Is the dependence mainly due to a shared regional component, a spatial process, or both?
4. How many regions are available for estimating this dependence?

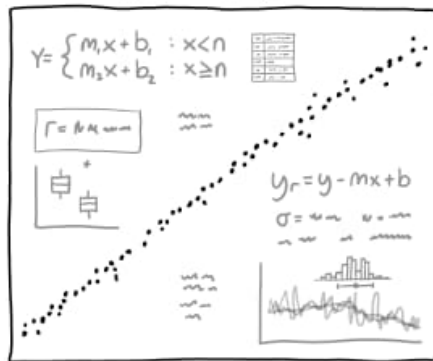
An important complication

Treating regions as independent clusters handles within-region dependence only if different regions can reasonably be regarded as independent.

If spatial dependence remains between regions, a standard cluster-robust analysis by region does not automatically account for it.

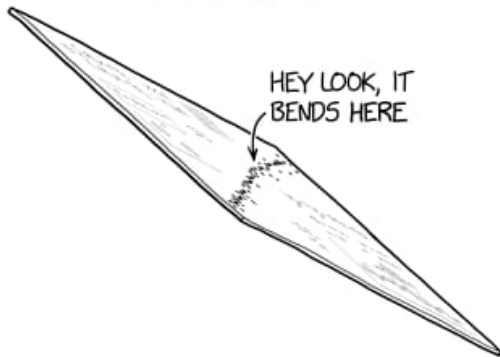
HOW TO DETECT A CHANGE IN THE SLOPE OF YOUR DATA

NOVICE METHOD:



DO A BUNCH OF STATISTICS

EXPERT METHOD:



TIP THE GRAPH SIDWAYS

References: generalized linear models

- Nelder, J. A. and Wedderburn, R. W. M. (1972). Generalized linear models. *Journal of the Royal Statistical Society: Series A*, 135(3), 370–384.
- McCullagh, P. and Nelder, J. A. (1989). *Generalized Linear Models*, 2nd ed. Chapman & Hall.
- Agresti, A. (2013). *Categorical Data Analysis*, 3rd ed. Wiley.
- Diggle, P. J., Heagerty, P., Liang, K.-Y. and Zeger, S. L. (2002). *Analysis of Longitudinal Data*, 2nd ed. Oxford University Press.
- Fitzmaurice, G. M., Laird, N. M. and Ware, J. H. (2004). *Applied Longitudinal Analysis*. Wiley.

References: dependence and robust covariance

- Huber, P. J. (1967). The behavior of maximum likelihood estimates under nonstandard conditions. In *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, Vol. 1, 221–233.
- White, H. (1980). A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica*, 48(4), 817–838.
- Liang, K.-Y. and Zeger, S. L. (1986). Longitudinal data analysis using generalized linear models. *Biometrika*, 73(1), 13–22.
- Cameron, A. C. and Miller, D. L. (2015). A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, 50(2), 317–372.
- Zeileis, A. (2006). Object-oriented computation of sandwich estimators. *Journal of Statistical Software*, 16(9), 1–16.
- Zeileis, A., Köll, S. and Graham, N. (2020). Various versatile variances: An object-oriented implementation of clustered covariances in R. *Journal of Statistical Software*, 95(1), 1–36.
- Cressie, N. (1993). *Statistics for Spatial Data*, revised ed. Wiley.

- Istituto Superiore di Sanità, EpiCentro. *Sorveglianza PASSI: rilevazione e campionamento.*
epicentro.iss.it/passi/procedure/rilevazione
- Istituto Superiore di Sanità, EpiCentro. *Sorveglianza PASSI: patologie croniche.*
epicentro.iss.it/passi/dati/croniche
- Istituto Superiore di Sanità, EpiCentro. *Sorveglianza PASSI.*
epicentro.iss.it/passi/